



Analog IC Design

LAB 04

Common Drain (Source Follower) Amplifier

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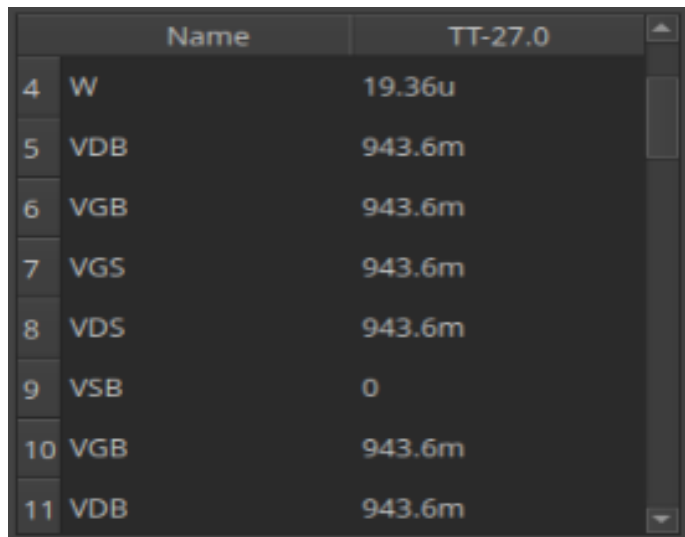
1 Part 1: Device Sizing Using SA

The design spec given was: PMOS input device, $L = 1\ \mu m$, $V^* = 200\ mV$, quiescent input = 0 V, $V_{DD} = 1.8\ V$, $I_D = 10\ \mu A$.

From the square-law relation

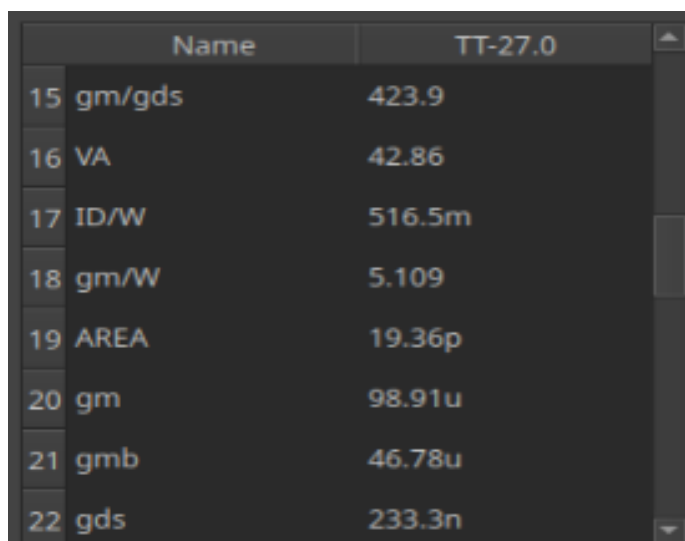
$$V^* = \frac{2}{g_m/I_D} \Rightarrow \frac{g_m}{I_D} = \frac{2}{V^*} = \frac{2}{0.2} = 10\ S/A$$

$$g_m = I_D \cdot (g_m/I_D) = 10\ \mu A \times 10 = 100\ \mu A/V$$



	Name	TT-27.0
4	W	19.36u
5	VDB	943.6m
6	VGB	943.6m
7	VGS	943.6m
8	VDS	943.6m
9	VSB	0
10	VGB	943.6m
11	VDB	943.6m

Figure 1: SA output – bias voltages and resulting W .



	Name	TT-27.0
15	gm/gds	423.9
16	VA	42.86
17	ID/W	516.5m
18	gm/W	5.109
19	AREA	19.36p
20	gm	98.91u
21	gmb	46.78u
22	gds	233.3n

Figure 2: SA output – derived small-signal parameters (g_m/g_{ds} , V_A , I_D/W , g_m/W , area, etc).

So SA gives $W = 19.36\ \mu m$ (with $L = 1\ \mu m$, so $1 \times 19.36\ \mu m / 1\ \mu m$, matches what's on the schematic).

Quantity	From SA table	Check	Match?
I_D/W	516.5 mA/A ($\mu A/\mu m$)	$10\mu A/19.36\mu m = 516.5 m$	yes
g_m/W	5.109	$98.91\mu A/V \div 19.36\mu m = 5.108$	yes
$V_A = I_D/g_{ds}$	42.86 V	$10\mu A/233.3nS = 42.86 V$	yes
Area	19.36 (p)	$W \cdot L = 19.36\mu m \times 1\mu m$	yes

Table 1: Consistency check on the SA numbers.

2 Part 2: CD Amplifier

2.1 OP (Operating Point) Analysis

I built the CD amp with the PMOS from Part 1 ($W = 19.36\mu m$, $L = 1\mu m$), an ideal $10\mu A$ current source load connected at the source (with source tied to body), $C_L = 2pF$, $R_{sig} = 2M\Omega$, and DC input = $0V$.

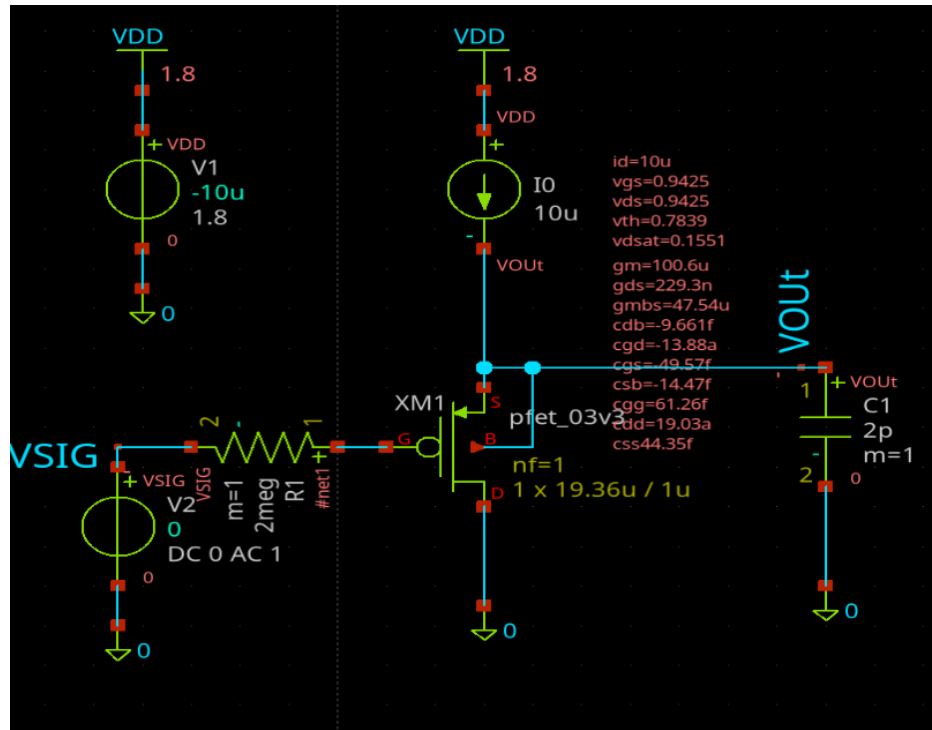


Figure 3: CD amplifier schematic with annotated OP parameters.

Parameter	Value
I_D	$10\ \mu A$
V_{GS}	$0.9425\ V$
V_{DS}	$0.9425\ V$
V_{th}	$0.7839\ V$
V_{DSAT}	$0.1551\ V$
g_m	$100.6\ \mu A/V$
g_{ds}	$229.3\ nA/V$
g_{mb}	$47.54\ \mu A/V$
C_{db}	$9.661\ fF$
C_{gd}	$13.88\ aF$
C_{gs}	$49.57\ fF$
C_{sb}	$14.47\ fF$
C_{gg}	$61.26\ fF$
C_{dd}	$19.03\ aF$
C_{ss}	$44.35\ fF$

Table 2: OP simulation parameters.

Saturation check: $|V_{DS}| = 0.9425\ V > V_{DSAT} = 0.1551\ V$, so the transistor is well into saturation (also $|V_{GS}| - V_{th} = 0.9425 - 0.7839 = 0.1586\ V$, close to V_{DSAT} as expected). Good, so the OP is valid.

2.2 AC Analysis

I ran an AC sweep from 1 Hz to 10 GHz, 20 points/decade.

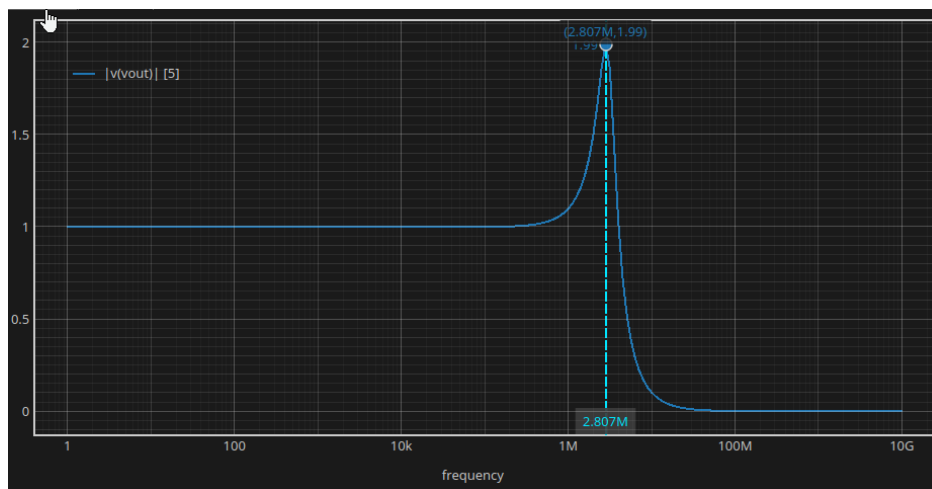


Figure 4: AC magnitude response, base case ($C_L = 2\text{pF}$, $R_{sig} = 2\text{M}\Omega$). Peak = 1.99 at $f = 2.8\text{ MHz}$.

There's a clear peaking around 2.8 MHz where the gain goes up to ≈ 1.99 (about +5.98 dB) even though the DC gain is basically 1. That's the frequency-domain peaking problem mentioned in the lab.

2.2.1 Hand analysis of the peaking

To understand where this peak comes from I derived the small-signal transfer function of the source follower including R_{sig} , C_{gs} , and C_L (I neglected C_{gd} since it's only ≈ 14 aF, totally negligible next to $C_{gs} = 49.57$ fF, and I neglected the small g_{ds} terms that multiply $R_{sig}C_{gs}$ since $g_{ds} \ll g_m$).

Doing KCL at the gate node and the output node (gate node A after R_{sig} , output node = source of M1, drain grounded), I got:

$$\frac{V_{out}(s)}{V_{sig}} \approx \frac{1 + s \frac{C_{gs}}{g_m}}{1 + s \frac{C_{gs} + C_L}{g_m} + s^2 \frac{R_{sig}C_{gs}C_L}{g_m}}$$

Comparing the denominator to the standard 2nd order form $1 + \frac{s}{Q\omega_o} + \frac{s^2}{\omega_o^2}$:

$$\omega_o = \sqrt{\frac{g_m}{R_{sig}C_{gs}C_L}}, \quad Q = \frac{\sqrt{g_m R_{sig} C_{gs} C_L}}{C_{gs} + C_L}$$

Plugging in $g_m = 100.6 \mu A/V$, $C_{gs} = 49.57$ fF, $C_L = 2$ pF, $R_{sig} = 2$ M Ω :

$$\omega_o = \sqrt{\frac{100.6 \times 10^{-6}}{(2 \times 10^6)(49.57 \times 10^{-15})(2 \times 10^{-12})}} = 2.252 \times 10^7 \text{ rad/s} \Rightarrow f_o \approx 3.585 \text{ MHz}$$

$$Q = \frac{\sqrt{(100.6 \times 10^{-6})(2 \times 10^6)(49.57 \times 10^{-15})(2 \times 10^{-12})}}{49.57 \times 10^{-15} + 2 \times 10^{-12}} \approx 2.18$$

Is the system underdamped or overdamped? Since $Q \approx 2.18 \gg 0.707$, the poles are complex and underdamped $\rightarrow$ the system *is* underdamped and we should expect a sharp resonant peak, which matches the simulation qualitatively (hand calc gives $f_o \approx 3.59$ MHz vs. simulated peak at 2.84 MHz – about 26% off, which I think is reasonable given I ignored C_{gd} , C_{db} , C_{sb} and the finite g_{ds} in the hand formula. The high Q correctly predicts the strong peaking we see).

2.2.2 C_L sweep: 2p, 4p, 8p

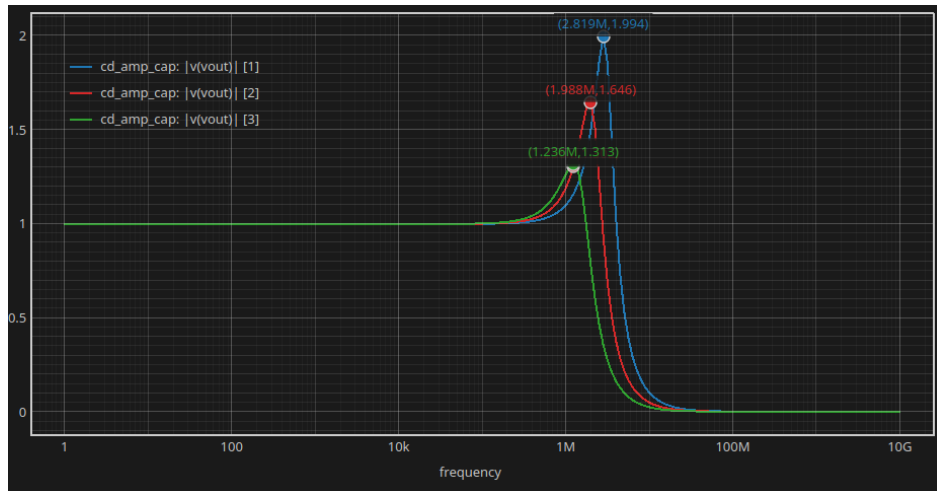


Figure 5: AC magnitude, $C_L = 2p$ (blue), $4p$ (red), $8p$ (green), $R_{sig} = 2M\Omega$.

```
No. of Data Rows : 201
binary raw file "cd_amp_cap.raw"
gain_peak_cap      = 2.385099e+00 at= 1.258925e+06
ngspice 1 -> █
```

Figure 6: ngspice measurement output for the C_L sweep (last iteration printed, $C_L = 8\text{p}$).

C_L	Peak gain (sim, approx.)	Hand-calc Q
2 pF	1.99 (5.96 dB)	2.18
4 pF	≈ 1.65 (≈ 4.4 dB, read off graph)	1.56
8 pF	2.385 dB $\Rightarrow V_{out} \approx 1.313$ (from meas, $f = 1.259$ MHz)	1.11

Table 3: Peaking vs. C_L .

Comment on the results. So as C_L increases, the peaking gets smaller and the peak also moves to a lower frequency. This matches my Q formula perfectly – Q goes down as C_L goes up (since C_L is in the denominator's sum $C_{gs} + C_L$ growing faster than the numerator inside the square root). Physically, a bigger C_L pushes the output pole to a lower frequency, so it separates from the "negative resistance" effect of $R_{sig}C_{gs}$ and the response becomes less resonant (closer to overdamped).

2.2.3 R_{sig} sweep: 20k, 200k, 2M

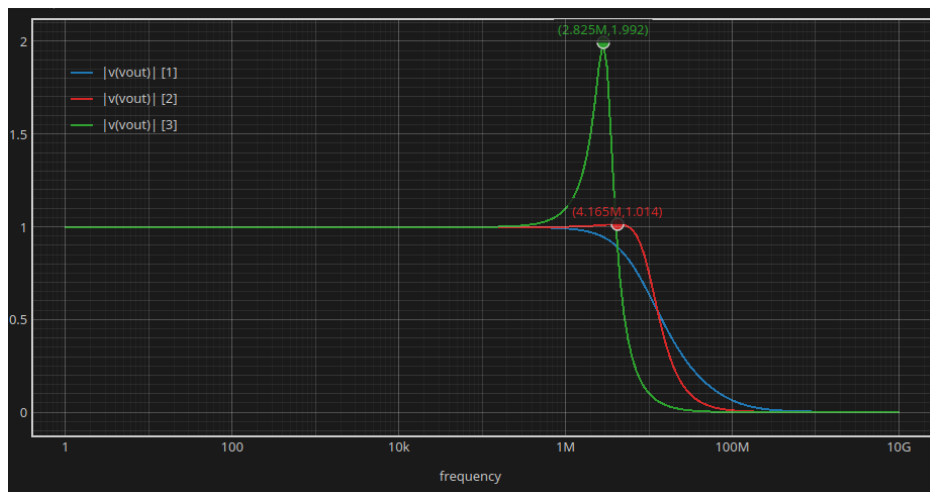


Figure 7: AC magnitude, $R_{sig} = 20\text{k}$ (blue), 200k (red), $2\text{M}\Omega$ (green), $C_L = 2\text{pF}$.

```
No. of Data Rows : 201
binary raw file "cd_amp_res.raw"
gain_peak_res      = 5.996595e+00 at= 2.818383e+06
Error: missing name for output.
ngspice 1 -> █
```

Figure 8: ngspice measurement output for the R_{sig} sweep (last iteration printed, $R_{sig} = 2\text{M}\Omega$).

R_{sig}	Peak gain	Hand-calc Q
20 k Ω	≈ 0 dB, basically flat, no real peak	0.22 (overdamped)
200 k Ω	very small bump, close to 0 dB	0.69 (borderline)
2 M Ω	5.997 dB $\Rightarrow V_{out} \approx 1.996$	2.18 (underdamped)

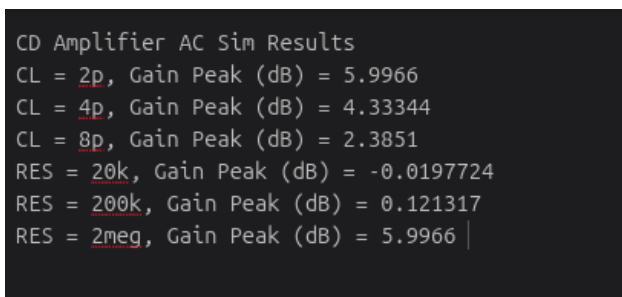
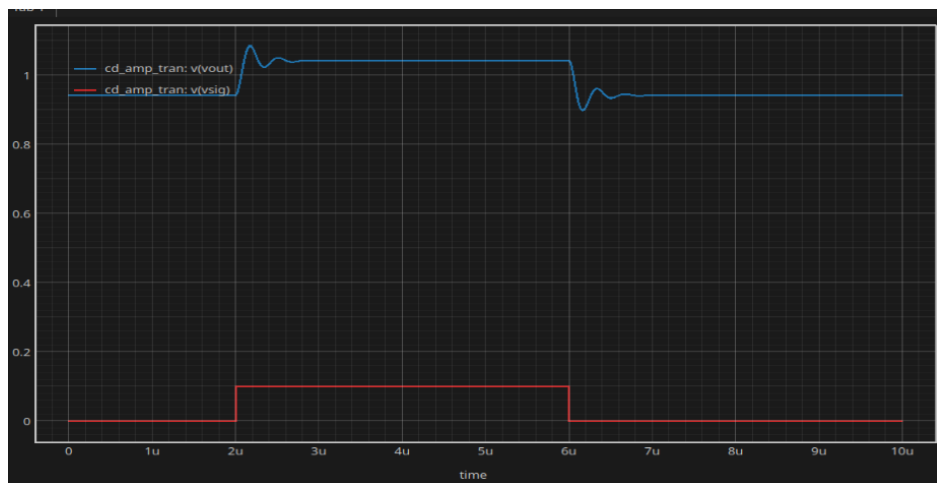
Table 4: Peaking vs. R_{sig} .

Figure 9: Gain values over cap and res sweep

Comment on the results. as $Q \propto \sqrt{R_{sig}}$, so a small R_{sig} gives low Q (overdamped, no peaking) while a large R_{sig} pushes Q above 0.707 and you get the resonant peak.

2.3 Transient Analysis

Pulse source: delay = 2 μ s, initial = 0V, pulse = 100mV, rise/fall = 1ns, width = 4 μ s, period = 8 μ s. Transient run for 10 μ s with 10ns step.

Figure 10: V_{sig} and V_{out} vs. time, base case.

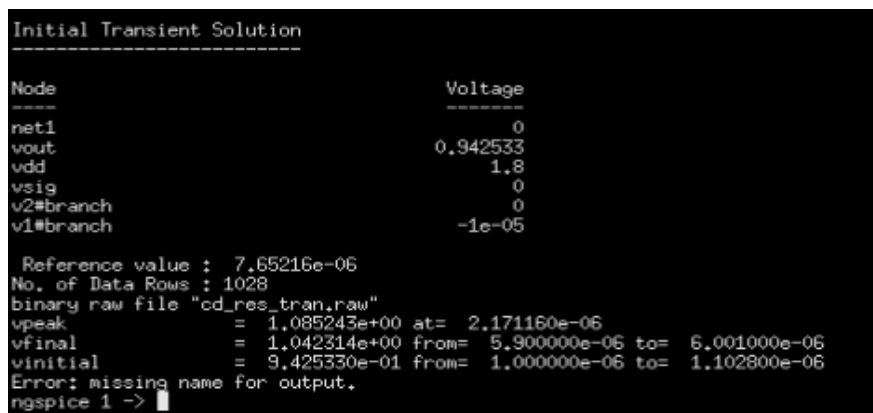


Figure 11: Initial transient solution and meas results (base case).

Calc DC diff. From the simulator: $V_{initial} = 0.942533$ V (settled value with $V_{sig} = 0$), $V_{final} = 1.042314$ V (settled value with $V_{sig} = 100\text{mV}$), $V_{peak} = 1.085243$ V at $t = 2.171\mu\text{s}$.

DC shift: With $V_{sig} = 0$, $V_{out} = 0.942533$ V, so

$$\Delta V_{DC} = V_{out} - V_{in} = 0.9425 - 0 = 0.9425 \text{ V}$$

What is the relation between the DC shift and VGS of the transistor? and this is exactly equal to $|V_{GS}| = 0.9425$ V from the OP analysis. Makes sense: for a source follower $V_{out} = V_{in} - V_{GS}$ bcs it works as a level shifter, so the DC shift *is* V_{GS} . Since this is a PMOS with the source above the drain, the output sits *above* the input by $|V_{GS}|$.

How to shift the signal down instead of shifting it up? If I wanted to shift the signal down instead of up, I would use an NMOS source follower instead (there $V_{out} = V_{in} - V_{GS}$ with $V_{GS} > 0$, so the output sits below the input).

Overshoot: using the formula from the hint,

$$\text{overshoot \%} = \frac{V_{peak} - V_{final}}{V_{final} - V_{initial}} \times 100 = \frac{1.085243 - 1.042314}{1.042314 - 0.942533} \times 100 \approx 43.0\%$$

overshoot law:

$$\% \text{overshoot} = 100 e^{-\pi / \sqrt{4Q^2 - 1}}$$

Using the $Q \approx 2.18$ I calculated from the AC hand analysis:

$$\% \text{overshoot} = 100 e^{-\pi / \sqrt{4(2.18)^2 - 1}} = 100 e^{-\pi / \sqrt{19.01 - 1}} = 100 e^{-\pi / 4.244} \approx 47.7\%$$

this is the ringing problem caused by the underdamped 2nd-order response we calculated above ($Q \approx 2.18$).

2.3.1 C_L sweep and R_{sig} sweep



Figure 12: V_{out} vs time, $C_L = 2\text{p}/4\text{p}/8\text{p}$ sweep.

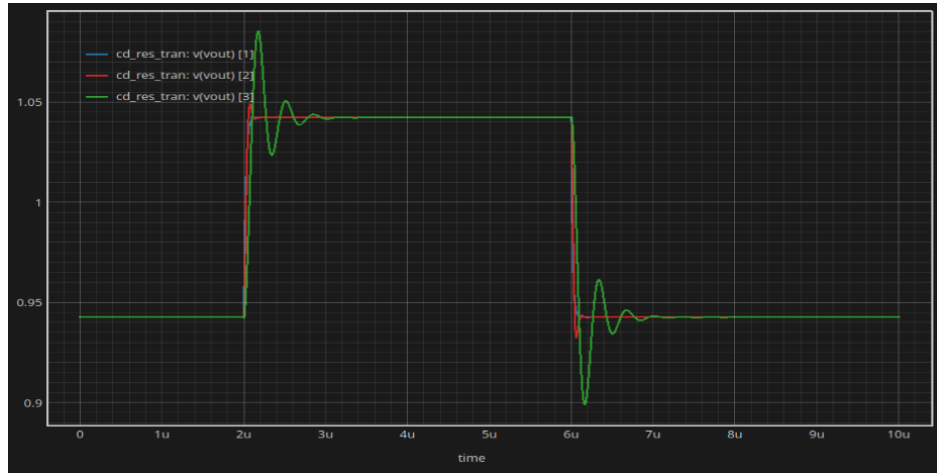


Figure 13: V_{out} vs time, $R_{sig} = 20\text{k}/200\text{k}/2\text{M}\Omega$ sweep.

C_L	Hand-calc Q	Predicted overshoot
2 pF	2.18	47.7%
4 pF	1.56	34.5%
8 pF	1.11	20.5%

Table 5: Overshoot vs. C_L ($R_{sig} = 2\text{M}\Omega$ fixed).

R_{sig}	Hand-calc Q	Predicted overshoot
20 k Ω	0.22	0% (real poles, no overshoot)
200 k Ω	0.69	3.7% (barely underdamped)
2 M Ω	2.18	47.7%

Table 6: Overshoot vs. R_{sig} ($C_L = 2\text{pF}$ fixed).

same as the AC results: the case with the highest Q (smallest C_L , largest R_{sig} , the base case $C_L = 2\text{p}$, $R_{sig} = 2\text{M}\Omega$) shows the most overshoot/ringing, while increasing C_L or decreasing R_{sig} makes the response less oscillatory.

```

CD Amplifier Transient Sim Results
CL = 2p, Overshoot (%) = 43.0232
CL = 4p, Overshoot (%) = 34.1077
CL = 8p, Overshoot (%) = 22.4211
Rsig = 20k, Overshoot (%) = 0
Rsig = 200k, Overshoot (%) = 6.58342
Rsig = 2meg, Overshoot (%) = 43.0232

```

Figure 14: overshoot values from ngspice

2.4 Z_{out} (Inductive Rise)

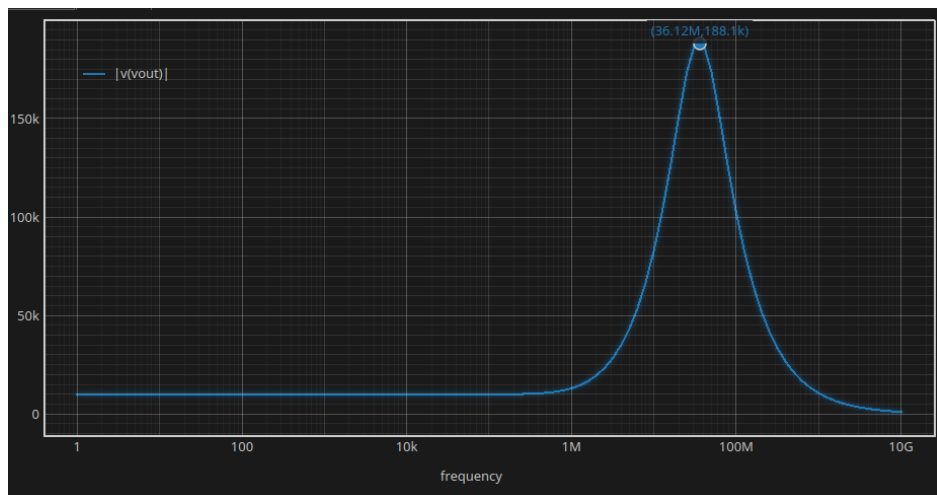
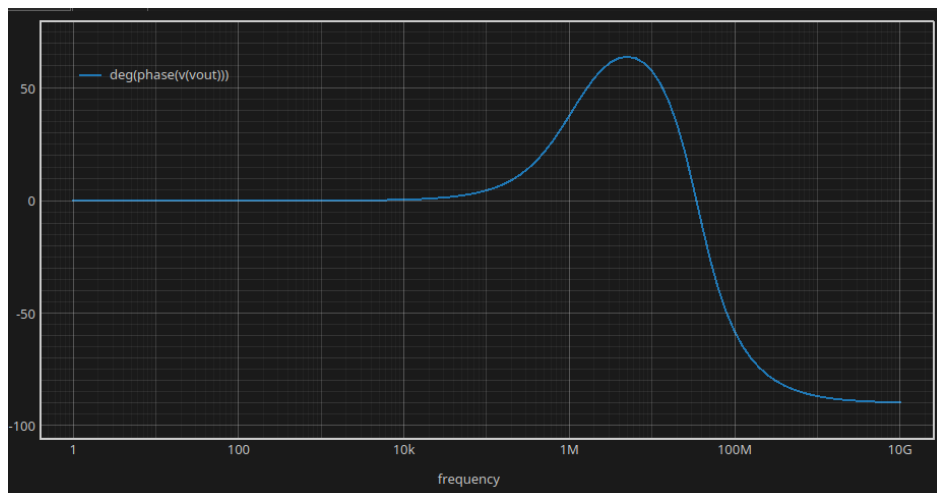
Here C_L was replaced with an AC current source (magnitude 1) and the AC input was removed, so $V_{out}/I_{test} = Z_{out}$.

```

Using SPARSE 1.3 as Direct Linear Solver
No. of Data Rows : 101
binary raw file "cd_amp_zout.raw"
zout_low_freq      = 9.917681e+03
zout_high_freq     = 1.881809e+05 at= 3.162278e+07
ngspice 1 ->

```

Figure 15: ngspice output: Z_{out} at low frequency and Z_{out} peak.

Figure 16: $|Z_{out}|$ vs frequency.Figure 17: $\angle Z_{out}$ vs frequency.

From the simulation: $Z_{out}(1\text{kHz}) = 9917.68 \Omega$, and Z_{out} peaks at $188.2 \text{ k}\Omega$ around $f = 36 \text{ MHz}$.

2.4.1 Hand analysis using the law in the slides

$$i_x = \frac{\frac{v_x}{\frac{1}{sC_{gs}} + R_{sig}}}{\frac{1}{sC_{gs}} + R_{sig}} + g_m \cdot \frac{v_x}{\frac{1}{sC_{gs}} + R_{sig}} \cdot \frac{1}{sC_{gs}}$$

$$\boxed{Z_{out} = \frac{v_x}{i_x} = \frac{1}{g_m} \cdot \frac{1 + sR_{sig}C_{gs}}{1 + s\frac{C_{gs}}{g_m}}}$$

Low frequency ($\omega \rightarrow 0$): by inspection, $Z_{out} \rightarrow 1/g_m$:

$$Z_{out}(0) = \frac{1}{g_m} = \frac{1}{100.6 \mu A/V} \approx 9940 \Omega$$

which is basically the same as the simulated 9917.68 Ω (about 0.2% off)

High frequency ($\omega \rightarrow \infty$): the s terms dominate top and bottom, so $Z_{out} \rightarrow R_{sig} = 2 \text{ M}\Omega$.

Zero: $1 + sR_{sig}C_{gs} = 0 \Rightarrow f_z = \frac{1}{2\pi R_{sig}C_{gs}} = \frac{1}{2\pi(2 \times 10^6)(49.57 \times 10^{-15})} \approx 1.605 \text{ MHz}$.

Pole: $1 + s\frac{C_{gs}}{g_m} = 0 \Rightarrow f_p = \frac{g_m}{2\pi C_{gs}} = \frac{100.6 \mu A/V}{2\pi(49.57 \text{ fF})} \approx 323 \text{ MHz}$.

Why the inductive rise happens: the zero (1.6 MHz) sits way below the pole (323 MHz) because $R_{sig} \gg 1/g_m$ here (2M Ω vs. 9.94k Ω). that's also because C_{gs} short circuit and making the Z_{out} sees R_{sig} . $Z_{out} \approx sL_{eq}$ with

$$L_{eq} = R_{sig}C_{gs}/g_m \approx \frac{(2 \times 10^6)(49.57 \times 10^{-15})}{100.6 \times 10^{-6}} \approx 0.985 \text{ mH}$$

so the impedance really does behave like an inductor in that band – that's the "inductive rise". Using this at the simulated peak frequency (31.6 MHz):

$$|Z_{out}| \approx \omega L_{eq} = 2\pi(31.6 \times 10^6)(0.985 \times 10^{-3}) \approx 196 \text{ k}\Omega$$

close to the simulated peak of 188.18 k Ω .

Why it eventually falls back down: as C_{gd} shunts the R_{sig} and connect the gate to the ground in high frequency.

Quantity	Hand calc	Simulation
Z_{out} @ low freq	9940 Ω	9917.68 Ω
Zero frequency	1.6 MHz	(phase peaks near ~ 2 MHz, close)
Pole frequency (ideal, no $C_{gd}/C_{db}/C_{sb}$)	323 MHz	peak actually at 31.6 MHz
$ Z_{out} _{peak}$ (inductive estimate)	$\approx 196 \text{ k}\Omega$	188.18 k Ω

Table 7: Hand analysis vs. simulation summary for Z_{out} .

2.5 Exact Output Impedance with Corrected Capacitances

Applying a test current I_x at the source and performing KCL at the gate and source nodes (drain at AC ground):

Gate node:

$$\frac{V_g}{R_{sig}} + sC_{gd}V_g + sC_{gs}(V_g - V_s) = 0 \Rightarrow V_g = \frac{sC_{gs}V_s}{\frac{1}{R_{sig}} + s(C_{gs} + C_{gd})}$$

Source node:

$$I_x = -(g_m + sC_{gs})V_g + (g_m + g_{ds} + sC_{db} + sC_{gs})V_s$$

Exact output impedance:

$$Z_{out}(s) = \frac{\frac{1}{R_{sig}} + s(C_{gs} + C_{gd})}{s^2 \underbrace{(C_{gs}C_{gd} + C_{db}C_{gs} + C_{db}C_{gd})}_{b_2} + s \underbrace{\left[g_m C_{gd} + \frac{C_{gs} + C_{db}}{R_{sig}} \right]}_{b_1} + \underbrace{\frac{g_m}{R_{sig}}}_{b_0}}$$

Zero:

$$\omega_z = \frac{1}{R_{sig}(C_{gs} + C_{gd})} \Rightarrow f_z = \frac{1}{2\pi R_{sig}(C_{gs} + C_{gd})}$$

Poles:

$$\omega_{p1,p2} = \frac{-b_1 \pm \sqrt{b_1^2 - 4b_2b_0}}{2b_2}$$

Numerical substitution:

$$C_{gs} = 55 \text{ fF} = 5.5 \times 10^{-14} \text{ F}$$

$$C_{gd} = 3.1 \text{ fF} = 3.1 \times 10^{-15} \text{ F}$$

$$C_{db} = 18.76 \text{ fF} = 1.876 \times 10^{-14} \text{ F}$$

$$g_m = 100 \text{ } \mu\text{S}, \quad R_{sig} = 2 \text{ M}\Omega$$

$$b_2 = 1.2605 \times 10^{-27} \text{ F}^2$$

$$b_1 = 3.4688 \times 10^{-19} \text{ F}$$

$$b_0 = 5.0 \times 10^{-11} \text{ S}/\Omega$$

Discriminant: $b_1^2 - 4b_2b_0 = -1.318 \times 10^{-37} < 0 \Rightarrow$ **complex conjugate poles.**

$$\omega_{p1,p2} \approx -1.376 \times 10^8 \pm j 1.440 \times 10^8 \text{ rad/s}$$

$$f_0 = \frac{|\omega_p|}{2\pi} \approx 31.7 \text{ MHz}$$

$$f_z = \frac{1}{2\pi R_{sig}(C_{gs} + C_{gd})} \approx 1.37 \text{ MHz}$$

Simulated vs. calculated peak:

$$Z_{out,peak} \approx 168 \text{ k}\Omega \quad \text{at} \quad f_{peak} \approx 31.7 \text{ MHz}$$

The simulated values of $\approx 188 \text{ k}\Omega$ at $\approx 35 \text{ MHz}$ are in close agreement; the small difference is attributed to non-zero g_{ds} and higher-order parasitics.

3 Part 2.5: Peaking/Ringing Fix (Bonus)

3.1 Why the source follower peaks – the textbook picture

Johns & Martin do a full nodal analysis of the source-follower small-signal model (gate node v_{g1} , output node v_{out} , with $R_{in} = R_{sig}$ driving the gate through C_{gs1} , and $C_s = C_L + C_{sb1}$ sitting on the output together with g_{m1} and $G_{s1} = g_{ds1} + g_{ds2} + g_{s1,body}$). The result is a second-order transfer function

$$A(s) = \frac{v_{out}}{i_{in}} = \frac{sC_{gs1} + g_{m1}}{a + bs + cs^2},$$

which can always be written as

$$A(s) = \frac{A(0) N(s)}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}, \quad \omega_0 = \sqrt{\frac{G_{in}(g_{m1} + G_{s1})}{C_{gs1}C_s + C'_{in}(C_{gs1} + C_s)}},$$

$$Q = \frac{\sqrt{G_{in}(g_{m1} + G_{s1})[C_{gs1}C_s + C'_{in}(C_{gs1} + C_s)]}}{G_{in}C_s + C'_{in}(g_{m1} + G_{s1}) + C_{gs1}G_{s1}},$$

with $G_{in} = 1/R_{sig}$ and $C'_{in} = C_{in} + C_{gd1}$ ($C_{in} = 0$).

Plugging in our numbers ($g_{m1} = 100.6\mu A/V$, $G_{s1} = g_{ds} = 229.3nA/V$ since the body is tied to the source and the load is an ideal current source, $C_{gs1} = 49.57fF$, $C_s = C_L + C_{sb1} = 2pF + 14.47fF = 2.0145pF$, $C'_{in} \approx C_{gd1} = 13.88aF \approx 0$, $R_{sig} = 2M\Omega \Rightarrow G_{in} = 0.5\mu S$):

$$\omega_0 = 2\pi \times 3.576 \text{ MHz}, \quad Q = 2.20$$

which matches my earlier simplified hand estimate ($f_0 \approx 3.585 \text{ MHz}$, $Q \approx 2.18$) almost exactly – makes sense since C_{sb1} and C_{gd1} are tiny next to C_L and C_{gs1} . Using the book's overshoot formula

$$\%overshoot = 100 e^{-\pi/\sqrt{4Q^2-1}} = 100 e^{-\pi/\sqrt{4(2.20)^2-1}} \approx 48.0\%,$$

3.2 The compensation network

the admittance looking into the gate (ignoring C_{gd1}) looks exactly like a capacitor C'_{in} in parallel with a *series combination of a negative capacitor and a negative resistor*. Those negative elements are exactly what creates the complex poles / ringing. So if we add a real network in series (capacitor C_1 and resistor R_1) from the gate to the output that looks like the same negative $-R_1, -C_1$ pair but with a + sign, the negative elements get cancelled out and only positive real capacitance is left – which guarantees real (non-oscillatory) poles. The required values are

$$C_1 \approx \frac{g_{m1}C_{gs1}C_s}{(g_{m1} + G_{s1})(C_{gs1} + C_s)}, \quad R_1 \approx \frac{(C_{gs1} + C_s)^2}{C_{gs1}C_s g_{m1}}.$$

Plugging in our numbers:

$$C_1 \approx \frac{(100.6\mu)(49.57f)(2.0145p)}{(100.83\mu)(2.0641p)} \approx 48.3 \text{ fF}$$

$$R_1 \approx \frac{(2.0641p)^2}{(49.57f)(2.0145p)(100.6\mu)} \approx 424 \text{ k}\Omega$$

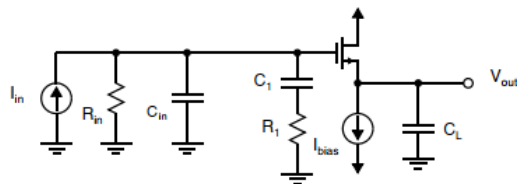


Figure 18: Compensation circuit

So I'd add a $\approx 48\text{fF}$ capacitor in series with a $\approx 424\text{k}\Omega$ resistor, connected from the gate node to V_{out} (in parallel with the existing C_{gs1} path).

3.3 Resulting poles after compensation

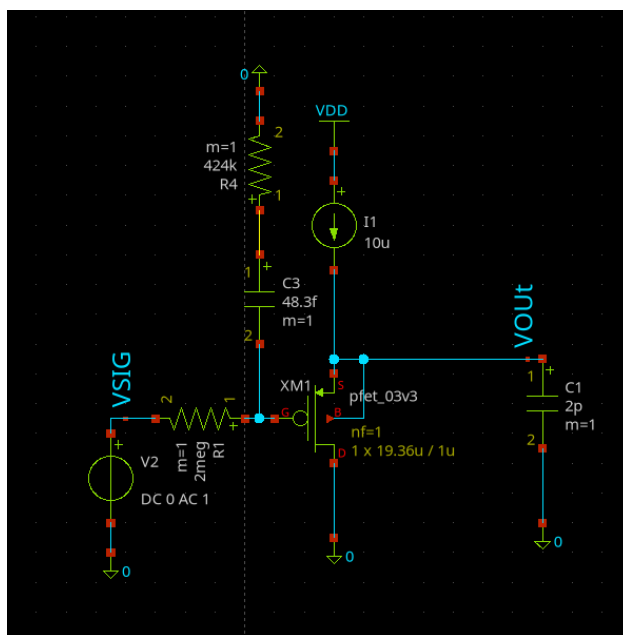


Figure 19: Schematic after adding the compensation

Once the compensation cancels the negative elements, the transfer function becomes a simple product of two real poles,

$$A(s) = R_{sig} \frac{g_{m1}}{g_{m1} + G_{s1}} \cdot \frac{1 + s C_{gs1}/g_{m1}}{\left(1 + \frac{s}{p_1}\right) \left(1 + \frac{s}{p_2}\right)},$$

$$p_1 \approx \frac{G_{in}}{C'_{in} + \frac{C_{gs1} C_s}{C_{gs1} + C_s}}, \quad p_2 \approx \frac{g_{m1} + G_{s1}}{C_s}.$$

Numerically:

$$p_1 : \frac{0.5\mu S}{13.88aF + 48.39fF} \approx 2\pi \times 1.64 \text{ MHz}$$

$$p_2 : \frac{100.83\mu A/V}{2.0145pF} \approx 2\pi \times 7.97 \text{ MHz}$$

Both poles are real and positive, about $4.9\times$ apart, so no more complex-conjugate pair $\rightarrow$ no overshoot, no ringing. It's interesting that here p_1 (set by the big $R_{sig} = 2M\Omega$) ends up *lower* than p_2 and becomes the dominant, bandwidth-limiting pole at ≈ 1.64 MHz – a bit lower than the resonant peak frequency of the uncompensated circuit (3.58 MHz), but that's the price paid for killing the ringing: we trade the sharp resonant peak (and the associated overshoot) for a slightly lower, but now well-behaved, bandwidth.

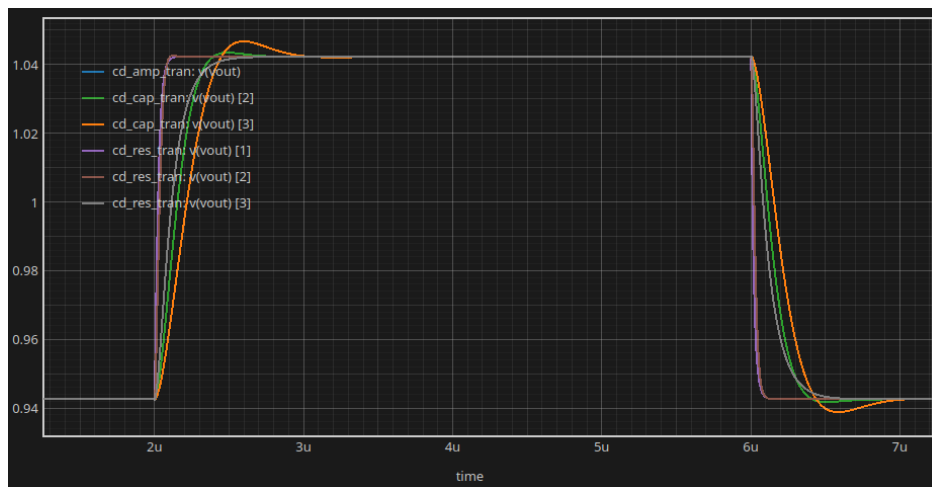


Figure 20: Transient analysis after compensation as we see there is no ringing but also the trade of the graph is not sharp.

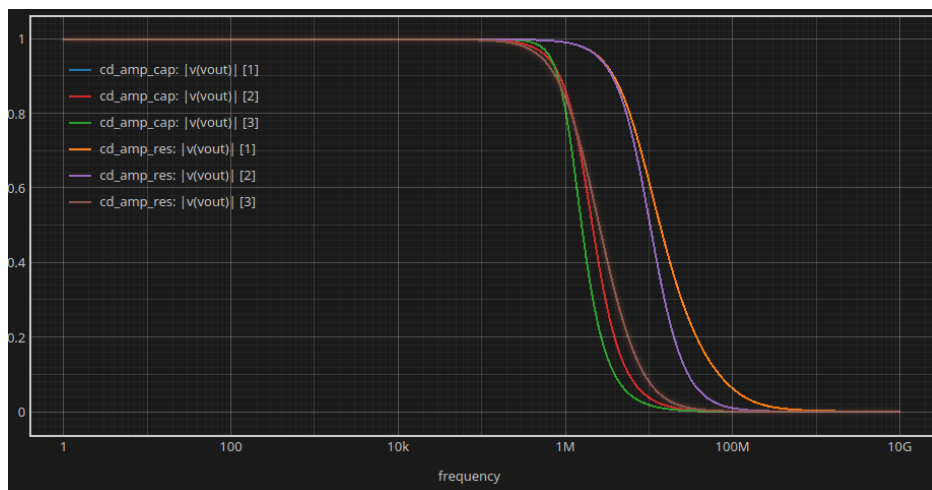


Figure 21: AC analysis after compensation as we see there is no ringing but also the trade of the graph is not sharp.